# Bookshelf

*A Reading Management Platform*

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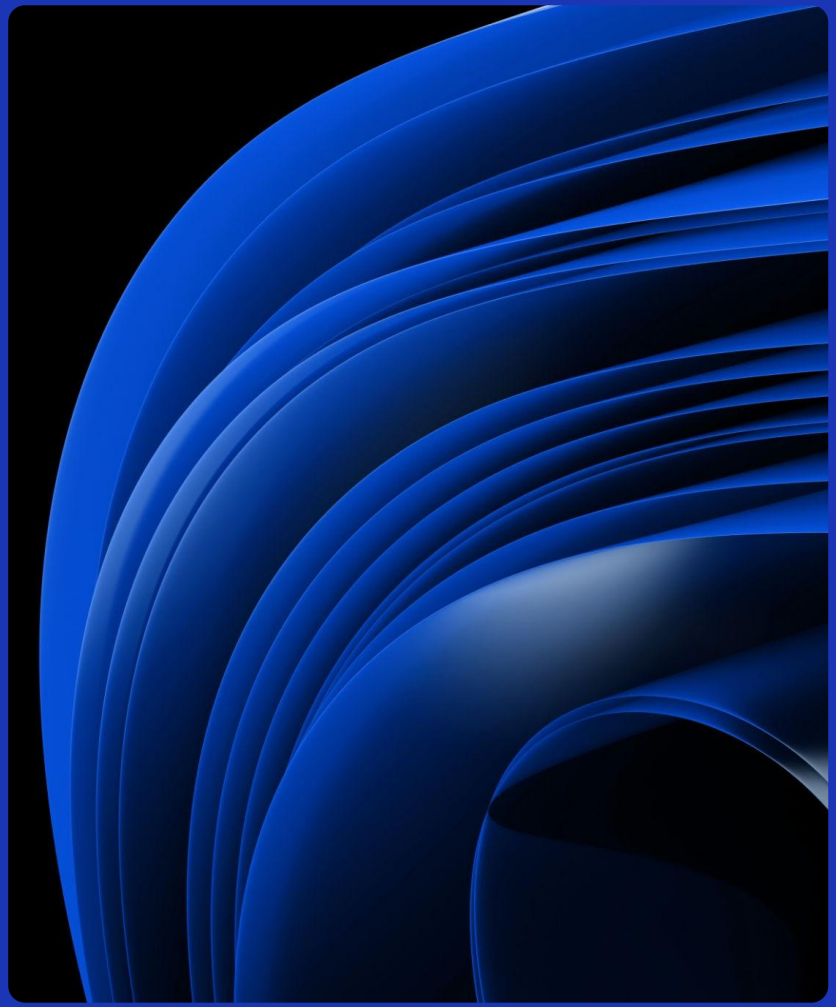

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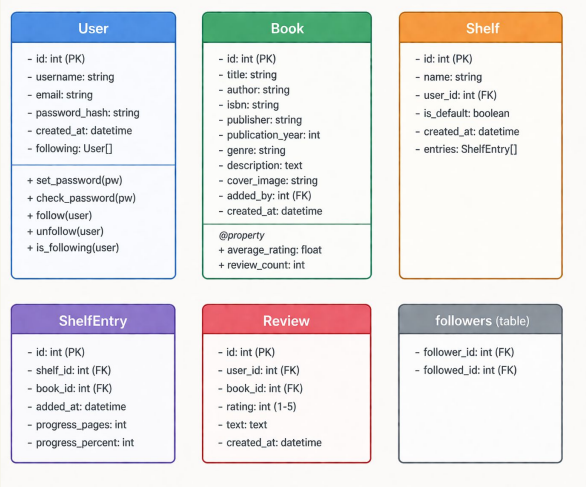
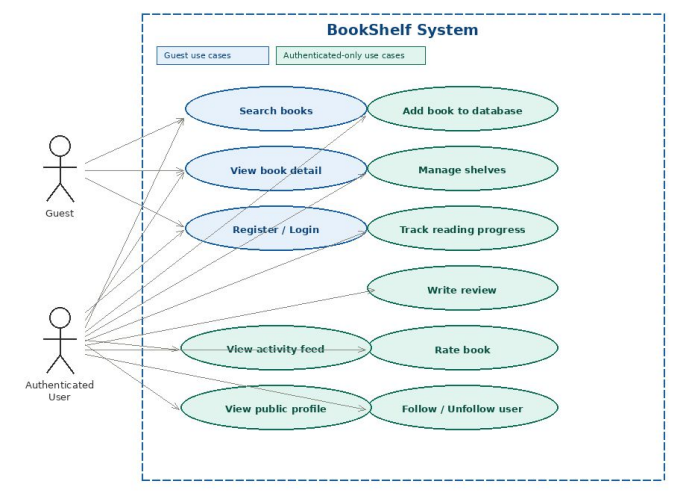
# What Bookshelf Does

- Discover and catalog books with detailed information
- Track reading progress across multiple shelves
- Write and share book reviews with star ratings
- Connect with other readers through social features
- Organize personal reading lists (Want to Read, Currently Reading, Finished)
- Search and browse a community-built book database

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# Project Goals

## Phase 1

### User friendly tracker

Users can understand how to use the website from signing up to uploading books and leaving reviews

## Phase 2

### Social reading group

Users can find and follow others on the platform and see all web activity such as new users joining, comments, or reviews

## Phase 3

### Secure authentication

Secure password hashing and minimum password requirements to ensure your data is protected

## Phase 4

### Responsive and Intiutive

The website responds to requests quickly and automatically updates when leaving a review, adding a book to a category or following users

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# Project Justification

## Personal Interest

- Combines my passion for reading and software development
- Oppertunity to build a beneficial and functional full-stack app for others
- Addresses a need for a good simple website to track books

## Technical Value

- Familiarized myself with web development tools (Flask, SQLAlchemy, HTML/CSS/JavaScript)
- Showcases database design with complex relationships (many-to-many, one-to-many)
- Applies security principles and best practices in a real-world application
- Develops skills in UI/UX design and user experience optimization

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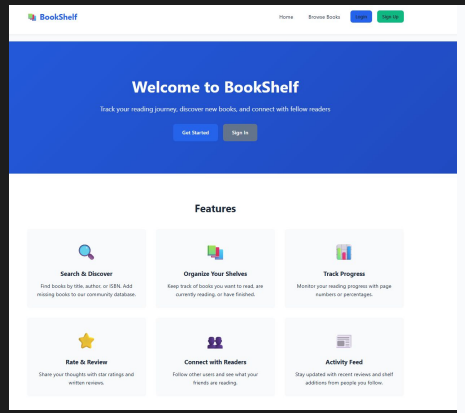
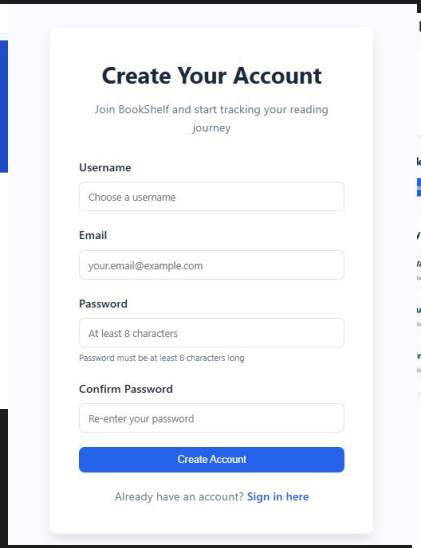
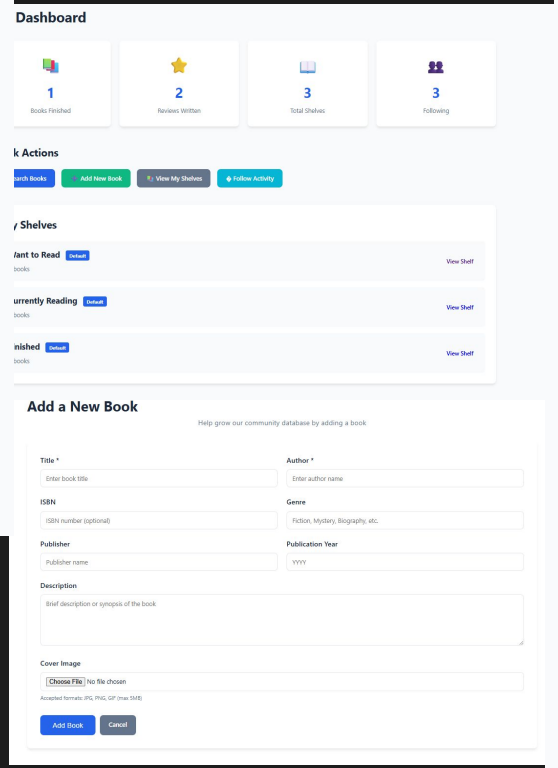
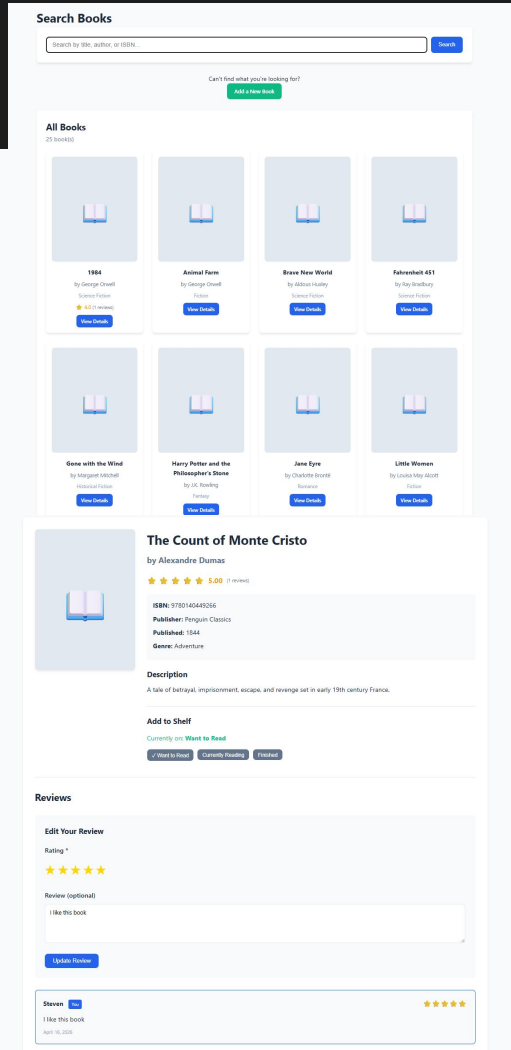
# How Bookshelf works

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# Live Demo

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# Project Challenges

## Database Relationships

Managing multiple many-to-many relationships (users following users, books on multiple shelves)

Implemented SQLAlchemy association tables with careful foreign key design and backref relationships

## Session Management and Security

Ensuring secure authentication without exposing passwords or allowing unauthorized access

Used Werkzeug password hashing with bcrypt, Flask session management, and login_required decorators

## File Uploads

Book cover images needed validation, secure storage, and size limits

Implemented server-side validation for file types/sizes, used secure_filename(), and created dedicated uploads directory

# Whats Left To Do

Add Cover Images for Books

Implement Profile/ Password Changes

Update Stylistic Choices

Add more detail to Book Descriptions

# If There Is Time

Add a book recommendation page

Add a book chat page

Add more books

Add a reply feature

# Questions?

## Thank you!

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